Minimum rival covers for 108 fixed RNA-Undesign motif instances

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Abstract

A collection of rival RNA structures can certify that a target is never the unique minimum-energy structure. We determine the smallest size of such a collection for 108 fixed motif instances from RNA-Undesign. With fixed boundaries and ViennaRNA's Turner 2004 model at 37 °C, the minima are between one and six, with a sum of 310 over the instances. Each optimum has an upper cover and a finite lower witness set. The accompanying checker reconstructs the candidate structures, checks lower bounds, and can exhaustively verify upper coverage. These are finite, model-specific results; they do not assert a bound for arbitrary RNA.

1 Instances, boundaries and energy model

The source is the pinned RNA-Undesign CSV, at `data/motifs_maxlen14_no_external/results.uniq.csv` in Git commit `8a8de22`. Of its 1,805 data rows, the 108 rows with `N_Rival` ≥ 2 are selected. The row identifiers in Tables **??**–**??** are one-based data-row numbers, excluding the header. RNA-Undesign supplies the motifs and the rival-based framework [**?**, **?**]; the task here is to minimize the number of rivals over an explicitly specified candidate set.

A motif has an outer base pair and zero or more boundary pairs denoted by `(*)`. Each boundary is an indivisible closed child. For energy evaluation, it is replaced by a hairpin with structure `(...)` and three fixed internal A letters. The outer and boundary pairs are preserved in every candidate. The internal A letters are never available as new pairing endpoints.

Let $\mathcal{X}(m)$ contain all sequences compatible with the target m: every target pair, including outer and boundary pairs, allows AU, UA, GC, CG, GU or UG; every other unfixed unpaired position allows A, C, G or U. Thus, with p target pairs and u unfixed unpaired positions,

$$|\mathcal{X}(m)| = 6^p 4^u.$$

Let $\mathcal{R}(m)$ be all noncrossing structures on the same motif positions that preserve the boundaries. A terminal hairpin must have at least three unpaired positions. The target itself is excluded. A boundary child counts as a closed child of its parent, rather than as an ordinary terminal hairpin in the motif grammar.

Energy $E(x, r)$ is evaluated using ViennaRNA 2.7.2, Turner 2004 parameters, 37 °C and `dangles=2` [**?**]. Special hairpins and isolated and GU pairs are allowed. Salt is 1.021 M; circular structures, G-quadruplexes and logarithmic multiloops are disabled. The integer `eval_structure_pt` API reports units of 0.01 kcal/mol. No floating-point tolerance is used in the cover comparison.

2 The finite minimum-cover claim

A rival $r \in \mathcal{R}(m)$ covers $x \in \mathcal{X}(m)$ if all its pairs are compatible with x and

$$E(x, r) \leq E(x, m).$$

Ties count: this prevents m from being the *unique* minimum-free-energy structure. The quantity determined is

$$k^*(m) = \min\big\{|C| : C \subseteq \mathcal{R}(m),\ \forall x \in \mathcal{X}(m)\ \exists r \in C :\ r \text{ covers } x\big\}.$$

An upper certificate is a list U_m of k distinct candidates covering all of $\mathcal{X}(m)$. A lower certificate is a nonempty finite set $W_m \subseteq \mathcal{X}(m)$ such that no set of at most $k-1$ candidates covers W_m.

Proposition 1. *If these upper and lower certificates hold, then $k^*(m) = k$.*

Proof. The upper certificate gives $k^*(m) \leq k$. A cover of $\mathcal{X}(m)$ also covers its subset W_m, so the lower certificate gives $k^*(m) \geq k$. For $k = 1$, the nonempty sequence space already excludes an empty cover. □

For a lower check, enumerate all candidates and evaluate their coverage on W_m. Represent each coverage set by a bit mask. Starting with $\{0\}$, take all unions with one candidate mask at each step through depth $k-1$. Absence of the full witness mask proves the lower bound. Reusing a candidate does not change a union, so the procedure includes every subset of at most $k-1$ candidates. No optimization solver or search implementation is required.

For an upper check, enumerate all $6^p 4^u$ sequences and require at least one compatible rival in U_m with energy no larger than the target. The candidates are regenerated by a stack grammar that treats boundaries atomically; the certificates do not supply a purportedly complete candidate list.

3 Results

The 108 minima and the candidate and sequence counts are listed in Tables **??**–**??**. There are 12,235 candidates in total across the instance-specific candidate sets, and 300,119,040 sequence assignments. The distribution is

Minimum k^*	1	2	3	4	5	6
Instances	10	31	40	19	6	2

The sum of the instance-specific minima is 310. It is not a claim of 310 distinct rivals forming one cover for all motifs. The lower witness sets contain one to six sequences; their size and the certificate byte size are not minimized. The original CSV's rival counts are outputs of its search procedure, not minimum-cardinality claims being corrected here.

4 Reproduction and scope

The package contains `data/results.csv`, `data/certificates.json`, the certificate checker and the source-fetch tool. The original CSV and its motif strings are not redistributed. A user obtains the pinned CSV separately; the tool accepts it only if its SHA-256 is

`a06d6cc495c36c50b4292e260ebc89d54d3dbd422b99fd2ebfae66431ff59041`.

Each certificate includes the SHA-256 of the exact decoded `Motif` field, encoded as UTF-8 without an added newline or normalization. Stored rivals use the compressed boundary notation; witness sequences use the expanded coordinates with fixed AAA. The checker requires the exact 108 selected row identifiers before reconstructing the motifs.

The default smoke run checks all candidate counts, sequence counts and lower certificates, and exhaustively checks upper covers for m0081, m0335 and m1780, totaling 539,136 sequences. The `--full` option checks every upper cover and all 300,119,040 sequences. The smoke run alone is not a full upper-coverage verification. The packaged checker derives from the research-side certificate grammar; this run is not a new blind independent review. The `VERIFICATION-RECORD.md` file separates the earlier complete independent certificate check from the subsequent package-assembly checks and records evaluator dependencies.

The mathematical claim concerns this fixed sequence space, candidate grammar, boundary treatment and parameter set. It does not cover unconstrained rival structures, other thermodynamic parameters, pseudoknots, measured molecular behavior or all RNA motifs. Computational certificate checks establish the stated finite claims relative to their implementation and energy evaluator; the software is not formally verified.

Rival-based certificates of undesignability and the enumeration of minimal undesignable motifs are prior work [**?**, **?**, **?**]. Minimality under inclusion of motifs is different from the minimum *number of rivals* for a fixed motif. Our contribution is the fixed collection of optimum values and their upper and lower certificates. No priority claim is made. A possible next question is how these minima change under a separately specified boundary or parameter change; the present certificates do not answer it.

AI systems assisted with the calculations, certificate checks, literature comparison, code and writing. Tatsuhiko Takimoto is responsible for the report. This report has not undergone human peer review.

References

[1] T. Zhou, W. Y. Tang, D. H. Mathews and L. Huang. Undesignable RNA Structure Identification via Rival Structure Generation and Structure Decomposition. RECOMB 2024. arXiv:2311.08339.

[2] T. Zhou, A. Malik, W. Y. Tang, D. H. Mathews and L. Huang. Scalable and Interpretable Identification of Minimal Undesignable RNA Structure Motifs with Rotational Invariance. RECOMB 2025. arXiv:2402.17206.

[3] T. Zhou, A. Malik, W. Y. Tang, D. H. Mathews and L. Huang. Theory, Algorithms, and Applications for Identification of Undesignable RNA Secondary Structures and Motifs. J. Comput. Biol. 33(1):134–167, 2026. doi:10.1177/15578666251398551.

[4] R. Lorenz et al. ViennaRNA Package 2.0. Algorithms Mol. Biol. 6:26, 2011. doi:10.1186/1748-7188-6-26.

Instance results (1/2)

ID	k^*	Candidates	Sequences
m0025	3	64	5,308,416
m0039	4	24	124,416
m0044	3	56	1,990,656
m0081	4	34	82,944
m0137	2	61	3,538,944
m0139	2	159	5,308,416
m0144	3	40	497,664
m0155	1	118	5,308,416
m0208	5	273	14,155,776
m0215	2	118	14,155,776
m0232	2	273	14,155,776
m0237	1	273	14,155,776
m0254	2	92	1,327,104
m0265	4	92	1,327,104
m0295	6	110	1,990,656
m0314	2	90	1,990,656
m0317	3	49	497,664
m0335	2	14	124,416
m0353	3	19	124,416
m0401	4	126	497,664
m0420	4	159	5,308,416
m0470	2	65	746,496
m0489	2	96	1,990,656
m0501	2	69	1,990,656
m0525	2	56	1,990,656
m0562	3	56	1,990,656
m0564	2	21	497,664
m0569	3	34	746,496
m0571	4	96	746,496
m0618	3	110	1,990,656
m0624	2	21	221,184
m0630	3	69	746,496
m0641	4	220	5,308,416
m0653	3	90	1,990,656
m0656	3	34	497,664
m0664	3	254	1,990,656
m0682	3	65	1,990,656
m0692	3	34	497,664
m0695	3	19	746,496
m0716	3	90	746,496
m0717	3	110	1,990,656
m0719	4	126	497,664
m0732	4	220	1,990,656
m0756	3	69	884,736
m0779	4	34	82,944
m0812	2	64	5,308,416
m0820	3	90	1,990,656
m0839	3	159	5,308,416
m0855	3	110	746,496
m0857	2	92	1,327,104
m0866	3	254	5,308,416
m0885	1	159	1,990,656
m0916	2	56	746,496
m0920	2	40	497,664

Table 1: First 54 selected source rows.

Instance results (2/2)

ID	k^*	Candidates	Sequences
m0941	1	92	5,308,416
m0968	2	14	746,496
m0973	2	34	1,990,656
m0998	2	159	5,308,416
m1006	3	118	1,990,656
m1016	4	126	1,327,104
m1022	2	65	1,990,656
m1053	3	92	5,308,416
m1056	2	118	5,308,416
m1067	4	220	1,990,656
m1075	2	126	1,327,104
m1082	1	61	1,327,104
m1086	4	110	746,496
m1099	5	126	1,327,104
m1142	3	90	746,496
m1180	4	126	497,664
m1208	3	69	1,990,656
m1213	4	220	1,990,656
m1218	4	90	746,496
m1220	2	90	746,496
m1230	3	24	124,416
m1271	2	159	5,308,416
m1293	3	110	1,990,656
m1385	5	273	5,308,416
m1401	3	118	1,990,656
m1419	3	96	746,496
m1448	3	90	746,496
m1473	3	96	1,990,656
m1491	3	45	3,538,944
m1495	2	118	5,308,416
m1508	2	64	1,990,656
m1522	2	34	497,664
m1528	4	254	1,990,656
m1602	6	132	1,327,104
m1616	3	90	1,990,656
m1619	5	273	14,155,776
m1626	2	65	1,990,656
m1629	3	65	1,990,656
m1647	1	159	1,990,656
m1662	5	273	14,155,776
m1663	3	254	1,990,656
m1664	3	254	1,990,656
m1668	3	92	5,308,416
m1680	3	132	3,538,944
m1702	2	159	5,308,416
m1704	1	92	1,327,104
m1718	4	126	497,664
m1740	4	273	5,308,416
m1760	1	159	1,990,656
m1780	1	34	331,776
m1791	5	132	3,538,944
m1797	1	132	3,538,944
m1802	3	254	1,990,656
m1803	2	132	3,538,944

Table 2: Remaining 54 selected source rows.